NTNU

DATA SCIENCE AND MACHINE LEARNING IN NATURAL SCIENCE - AUTUMN 2026

Final Project Report

Overall Instructions

> ! **Deadline: 4th of December 2026**
>
> Submit your final project report and required attachments by the stated deadline, through Inspera platform.

Submission Requirements:

- **Report format:** Use the Elsevier Astronomy and Computing journal template.
- **Length:** Maximum 8 pages, excluding references.
- **File format:** PDF.
- **Figures:** Include all figures in the report with appropriate captions.
- **References:** Properly cite all literature, datasets, software, and other sources used in the project.
- **Submission:** Submit the report and required attachments through Inspera.

> i **Purpose of the Final Report**
>
> The final report should present your machine-learning project as a complete scientific study. It should clearly describe the problem, data, methodology, experimental setup, results, limitations, and conclusions.
> Your results should be evaluated quantitatively and compared with an appropriate baseline. The discussion should demonstrate that you understand both what your model achieves and where it fails.

Required Content

1. Introduction

The introduction should establish the scientific context and clearly define the problem addressed in the project.

Include:

- Motivation and significance of the problem.
- A clear problem statement or research question.
- A concise review of relevant literature, including:
  - previous machine-learning approaches to the problem;
  - traditional or non-machine-learning approaches, where relevant;
  - the current state of the art and its limitations.

2. Methodology

2.1 Data Collection and Description

Describe the dataset in sufficient detail for the reader to understand what information is available and where it comes from.

Include:

- The data source and data collection procedure.
- Dataset size and structure.

- Description of the relevant variables/features.
- Data types and units, where applicable.
- Data source and collection date, where relevant.
- Definition of the target variable or labels.

2.2 Exploratory Data Analysis

Provide an appropriate exploratory analysis of the dataset before modelling.

Depending on the problem, this may include:

- Summary statistics for important variables (e.g., mean, median, standard deviation).
- Analysis of relevant data distributions.
- Missing-value analysis.
- Correlation analysis, where appropriate.
- Relevant visualizations of the data.

2.3 Feature Engineering and Data Preprocessing

Describe all important transformations applied before model training.

Include, where applicable:

- Missing-value handling.
- Data cleaning.
- Feature scaling or normalization.
- Feature selection.
- Creation or transformation of features.
- Encoding of categorical variables.
- Other preprocessing specific to your dataset.

Important preprocessing decisions should be justified.

2.4 Machine-Learning Approach

Describe the machine-learning methods used in the project.

- **Baseline model:** Describe and justify the baseline used to evaluate whether the main machine-learning approach provides an improvement.
- **Main model(s):** Describe the algorithm or architecture and justify why it is appropriate for the problem.
- **Hyperparameters:** Explain how important hyperparameters were selected or tuned.
- **Model complexity:** Discuss relevant choices concerning model complexity and regularization, where applicable.

2.5 Experimental Setup

Clearly describe how model performance was evaluated.

Include:

- Training, validation, and test set proportions.
- The splitting methodology used (e.g., random, grouped, temporal, spatial).
- Cross-validation, if used.
- Training procedure.
- Relevant software and computational tools.
- Hardware specifications where they materially affect the work.

The test set should remain independent of model training and model-selection decisions.

Results and Discussion

3. Model Performance

Report quantitative performance using metrics appropriate for your problem.

For example:

- **Regression:** MSE, RMSE, R^2, MAE, or other justified metrics.
- **Classification:** accuracy, precision, recall, F1-score, AUC, or other justified metrics.

Your results should include:

- Performance on training, validation, and test data, as appropriate.
- Comparison with the baseline.
- Statistical significance of results (if applicable)

4. Visual Analysis

Use visualizations that help diagnose and interpret model performance.

For regression tasks, appropriate plots may include:

- Parity plots (predicted versus observed values).
- Residual distributions.
- Residuals versus predicted values.
- Learning curves.

For classification tasks, appropriate plots may include:

- Confusion matrices.
- ROC curves and AUC values.
- Precision–Recall curves.

5. Discussion

Interpret the results. Your discussion should address:

- Interpretation of results.
- Model strengths and limitations.
- Comparison with existing approaches.
- Error analysis and potential improvements.

Conclusions

Summarize the main outcomes of the project. Include:

- Summary of key findings
- Practical implications
- Future work suggestions

Avoid introducing substantial new results or methodology in the conclusion.

Required Attachments

! Dataset Files

In addition to the final report, submit the datasets used for model development and evaluation:

- Training set.
- Validation set.
- Test set.

The submitted files should correspond to the data split described in your report.

i Sensitive or Restricted Data

If your dataset contains information that cannot be shared directly, you may appropriately scale or otherwise modify the data before submission. Any such modification should preserve the structure needed to understand the analysis and should be briefly documented.